# Dynamical Localization in the Quantum Kicked Rotor

How quantum mechanics stops chaotic diffusion

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# The question

- Classical chaos: nearby orbits separate like $e^{\lambda t}$, and a chaotic system can spread forever (diffusion).
- Quantum mechanics: the Schrödinger equation is linear and unitary. Nothing separates exponentially.
- So what does a quantum system do when its classical version is chaotic?

**The simplest test case: the kicked rotor.**

- Classically, strong kicks make the momentum perform a random walk: the energy grows linearly forever.
- Quantum mechanically, the growth follows the classical one for a while, then stops.
- This is dynamical localization. It is closely related to Anderson localization, and it was seen with cold atoms in 1995.

# The kicked rotor

A particle on a ring, kicked once per period with a strength that depends on its angle:

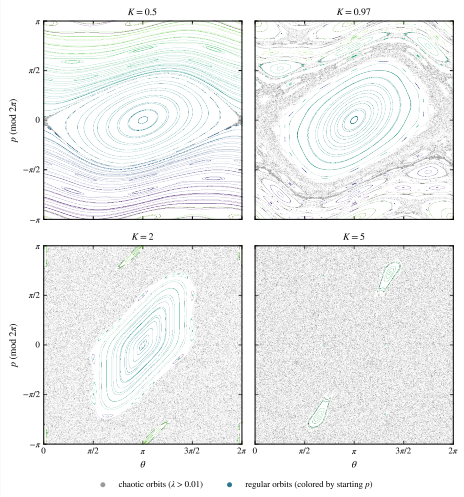
$$H = \frac{p^2}{2} + K\cos\theta \sum_n \delta(t-n)$$

Integrating Hamilton's equations over one period gives the standard map:

$$p_{n+1} = p_n + K\sin\theta_n, \qquad \theta_{n+1} = \theta_n + p_{n+1} \pmod{2\pi}$$

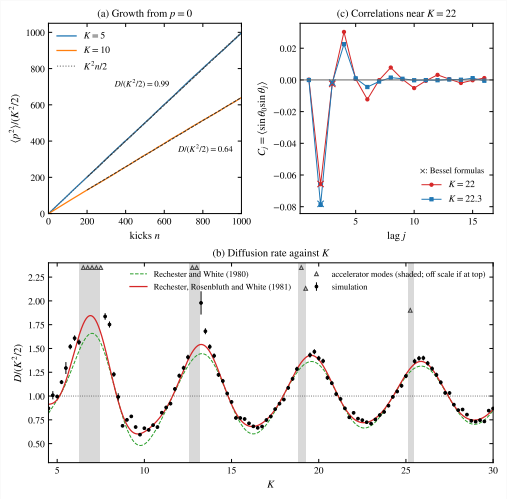
- $K$: kick strength, controls the chaos.
- The momentum is never wrapped: its growth is what we measure.
- Quantum version: $p = \hbar m$ with integer $m$; $\hbar$ is the effective Planck constant ($\hbar \to 0$ is the classical limit).
- Matched starting points: quantum state $m = 0$; classical ensemble $p = 0$, $\theta$ uniform.

# Classical dynamics: from order to chaos




- Gray: chaotic orbits ($\lambda > 0.01$). Colors: regular orbits.
- Weak kicks: invariant curves act as walls in momentum.
- The last wall breaks at $K_c = 0.9716$ [Greene 1979].
- Chaotic area: 1%, 43%, 76%, 98% at $K = 0.5$, 0.97, 2, 5.
- At $K = 5$: $\lambda = 0.98$, close to Chirikov's $\ln(K/2) = 0.92$ [Chirikov 1979].

# Classical diffusion and its Bessel corrections




- Random phases give $\langle p^2 \rangle \approx Dn$ with $D = K^2/2$.
- Correlations $C_j = \langle \sin\theta_0 \sin\theta_j \rangle$ correct it:

$$\frac{D}{K^2/2} \approx 1 - 2J_2 - 2J_1^2 + 2J_3^2 + 2J_2^2$$

[Rechester & White 1980; Rechester, Rosenbluth & White 1981]

- Measured: $D/(K^2/2) = 0.995 \pm 0.009$ at $K = 5$, $0.638 \pm 0.006$ at $K = 10$.
- Shaded: accelerator modes, where orbits fly ballistically.
- Remaining misses come from long correlations ($C_4$ at $K = 22$).

# The quantum rotor: one period at a time

The kick is instantaneous, so one period is exactly a kick followed by free rotation:

$$\hat{U} = \mathrm{e}^{-i\hat{p}^2/2\hbar}\,\mathrm{e}^{-iK\cos\hat{\theta}/\hbar}, \qquad \langle m'|\hat{U}|m\rangle = \mathrm{e}^{-i\hbar m'^2/2}\,(-i)^{m'-m}\,J_{m'-m}(K/\hbar)$$

- One kick couples each momentum state to about $k = K/\hbar$ neighbors.
- Two parameters: $K$ (classical chaos) and $\hbar$ (how quantum).

**Numerics:** store $\psi$ on $N$ angles; per period

1. multiply by $\mathrm{e}^{-iK\cos\theta/\hbar}$, 2. FFT to momentum, 3. multiply by $\mathrm{e}^{-i\hbar m^2/2}$, 4. inverse FFT.

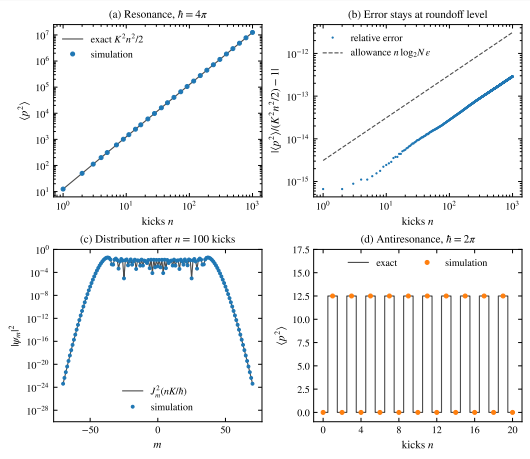
- Exact, not an approximation (kick and free step act at different times). Only the momentum range and roundoff limit it.
- $N = 2^{14}$ states; probability near the grid edge kept below $10^{-10}$. About 1 ms per kick.

# Validation: five checks, 39 tests

| Check | Test | Result |
|---|---|---|
| 1. Norm | $\langle\psi\|\psi\rangle = 1$ over 1000 kicks | drift $3 \times 10^{-16}$ per kick |
| 2. Resonance | $\hbar = 4\pi$: $\langle p^2\rangle = K^2n^2/2$; $\hbar = 2\pi$: return after 2 kicks | errors $\leq 3 \times 10^{-13}$ |
| 3. Early times | exact $\langle p^2\rangle$ after 1 and 2 kicks; $\hbar^2$ convergence at 3 kicks | $10^{-15}$; observed order 1.97 |
| 4. Grid | $N = 2^{14}$ vs $2^{15}$; edge probability | edge $3 \times 10^{-27}$; $2^{12}$ fails |
| 5. Time reversal | $n$ kicks forward, $n$ back | quantum $3 \times 10^{-13}$ after 1000; classical fails at 38 |

- Every tolerance has a stated reason: machine precision, statistical error, or known physics.
- Testing the tests: six planted bugs, each caught by at least one test.

# An exact test: quantum resonance




- $\hbar = 4\pi$: $\mathrm{e}^{-i\hbar m^2/2} = 1$, so the kicks add up coherently:

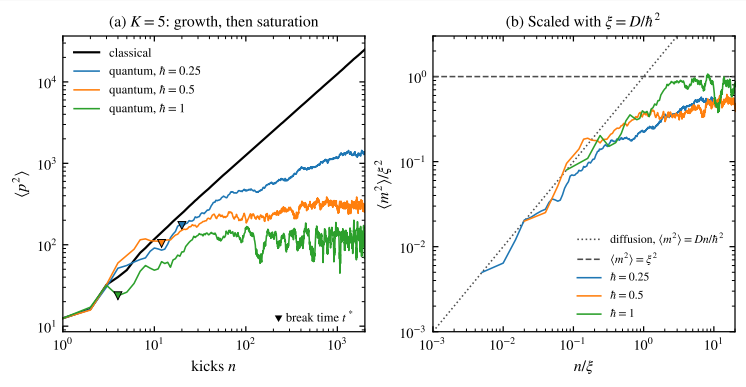
$$|\psi_m|^2 = J_m^2(nK/\hbar),$$

$$\langle p^2 \rangle = \tfrac{1}{2}K^2n^2$$

  Ballistic, faster than diffusion [Casati et al. 1979].

- $\hbar = 2\pi$: free step shifts $\theta$ by $\pi$, the next kick cancels: exact return every 2 kicks ("antiresonance").
- The code reproduces both to $\sim 10^{-13}$.

# Main result: the quantum growth stops




- Classical: $D = 12.51 \pm 0.10$, grows forever. Quantum ($K = 5$): follows, then saturates.
- Break times (25% below classical): 20, 12, 4 kicks at $\hbar = 0.25$, 0.5, 1. Right: rescaled with $\xi = D/\hbar^2$.

# Why it stops: Floquet theory

Eigenstates of the one-period operator: $\hat{U}|\phi_k\rangle = \mathrm{e}^{-i\omega_k}|\phi_k\rangle$ (quasienergies $\omega_k$). Then

$$\langle p^2\rangle_n = \sum_{k,k'} c_k^* c_{k'}\, \mathrm{e}^{in(\omega_k-\omega_{k'})}\, \langle\phi_k|\hat{p}^2|\phi_{k'}\rangle$$

- If the $|\phi_k\rangle$ are localized in momentum, only finitely many terms matter: the energy is quasi-periodic and bounded.

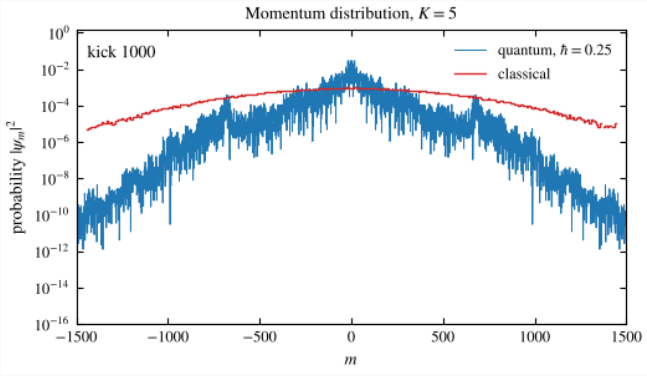
**How long until it stops?**

- About $\xi$ states involved, quasienergy spacing $\sim 2\pi/\xi$, resolved after $t^* \sim \xi$ kicks.
- Until then it diffuses: $\langle m^2\rangle \approx Dt/\hbar^2$. Consistency: $\xi^2 \sim D\xi/\hbar^2$, so

$$\xi \sim \frac{D}{\hbar^2}, \qquad t^* \sim \frac{D}{\hbar^2}, \qquad \langle p^2\rangle_\infty \sim \frac{D^2}{\hbar^2}$$

- The limits $\hbar \to 0$ and $n \to \infty$ do not commute.

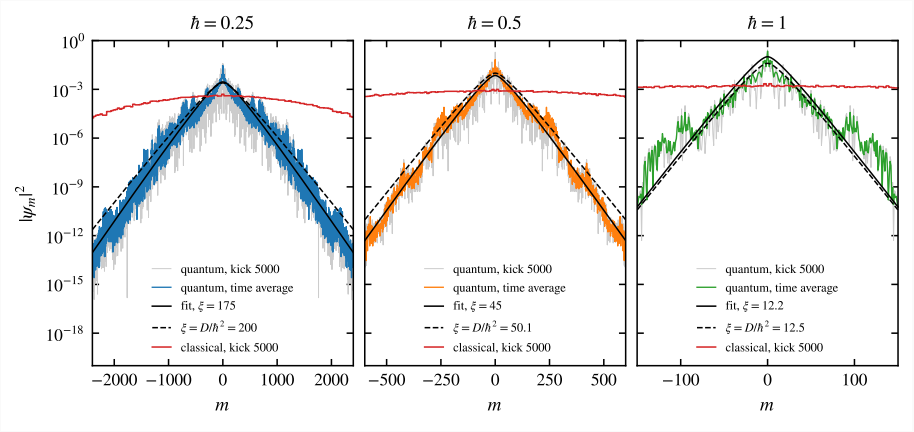
# Watching it happen




Momentum distribution, $K = 5$, $\hbar = 0.25$, up to 1000 kicks (frames speed up). Red: classical keeps widening. Blue: quantum stops.

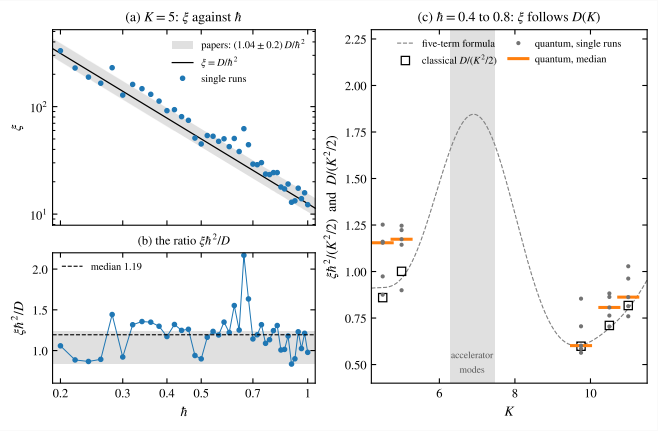
(The animation plays in Adobe Acrobat Reader, other viewers show the last frame; the file `figures/04_momentum_distribution.gif` plays anywhere.)

# The localized momentum distribution




- Exponential tails over 13 decades. Izrailev's shape [Izrailev 1990]: $\overline{P}(m) = \frac{1}{2\xi}(1 + \frac{2|m|}{\xi})e^{-2|m|/\xi}$
- Fitted $\xi/(D/\hbar^2) = 0.87, 0.90, 0.98$; earlier work: $1.04 \pm 0.20$ [Shepelyansky 1986].

# How the localization length scales




- 41 values of $\hbar$ at $K = 5$: $\xi$ follows $D/\hbar^2$ over a factor 27.
- $\xi\hbar^2/D$: median 1.19, spread 0.25; it wanders in bands (cause unknown).
- Varying $K$: $\xi$ follows the classical $D(K)$, including its Bessel dip at $K \approx 9.75$, not $K^2/2$.
- Near a rational $\hbar/4\pi$ localization can fail: $0.714/4\pi$ is $1.4 \times 10^{-7}$ from $5/88$.

# Why the states localize: the Anderson analogy

Fishman, Grempel and Prange [1982]: write the kick as $\mathrm{e}^{-ik\cos\theta} = \frac{1+iW}{1-iW}$, $W = -\tan\frac{k\cos\theta}{2}$. The Floquet equation becomes

$$T_m v_m + \sum_{r\neq 0} W_r\, v_{m+r} = -W_0\, v_m, \qquad T_m = \tan\Big(\frac{\omega - \hbar m^2/2}{2}\Big)$$

- A particle hopping on a 1D lattice of momentum states, with on-site energies $T_m$.
- $\hbar/4\pi$ irrational: $\hbar m^2/2 \bmod 2\pi$ looks random, $T_m$ is Cauchy distributed: the Lloyd model, all states localized.
- $\hbar/4\pi$ rational: $T_m$ periodic, Bloch waves, unbounded growth (resonance).
- Time-periodic kicking plays the role of static disorder.

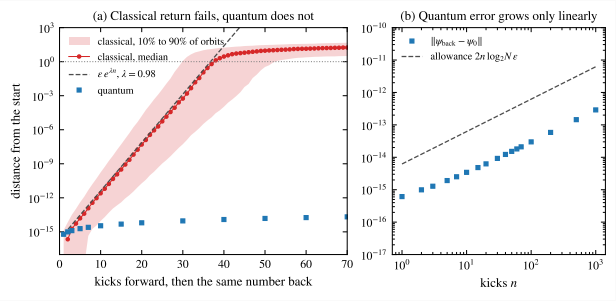
# Early times: how long does quantum follow classical?

After one kick both give $K^2/2$ exactly. After two kicks:

$$\langle p^2\rangle_2^{\mathrm{qm}} = K^2\Big[1 + \cos\tfrac{\hbar}{2}\big(J_0(K_q) - J_2(K_q)\big) - \tfrac{1}{2}J_2(K_2)\Big], \quad K_q = \tfrac{2K}{\hbar}\sin\tfrac{\hbar}{2},\ K_2 = \tfrac{2K}{\hbar}\sin\hbar$$

- $\hbar \to 0$ gives the classical formula; the gap is $O(\hbar^2)$: 1.0% at $\hbar = 0.25$.
- The $\hbar^2$ coefficient explodes: $-0.2$, $-35$, $-550$ at $n = 2, 3, 4$.
- Chaos magnifies quantum effects by $\mathrm{e}^{\lambda}$ per kick: detailed agreement only until the Ehrenfest time $t_E \sim \lambda^{-1}\ln(1/\hbar)$, a few kicks.
- After that only statistics survive (the diffusion rate), until the break time $t^* \sim D/\hbar^2$.

# Time reversal: chaos against unitarity




- Run $n$ kicks forward, $n$ back.
- Classical: roundoff grows like $\varepsilon\, e^{\lambda n}$; fails at $n = 38$ (predicted 36.7).
- Quantum: $\|\hat{U}(\psi + \delta) - \hat{U}\psi\| = \|\delta\|$, so errors only add up: $3 \times 10^{-13}$ after 1000 kicks.
- Bonus: the same code on Windows and Linux gives identical quantum numbers, but different classical orbits (statistics agree).

# Three time scales

| Time | Scale | What holds | $\hbar = 0.25$, $K = 5$ |
|---|---|---|---|
| Ehrenfest time | $t_E \sim \lambda^{-1} \ln(1/\hbar)$ | detailed correspondence | about 1 kick |
| Break time | $t^* \sim D/\hbar^2$ | classical diffusion rate | 20 kicks (25% rule) |
| Saturation | a few $\xi$ | localized, quasi-periodic | a few hundred kicks |

- As $\hbar \to 0$ at fixed $n$, both times go to infinity: the classical limit is recovered.
- At any fixed $\hbar$, the long-time behavior is fundamentally quantum.
- The saturated energy still swings slowly (17% rms over 40000 kicks): beats between quasienergies.

# In the lab: cold atoms

Moore, Robinson, Bharucha, Sundaram and Raizen [1995]:

- Ultracold sodium atoms in a pulsed standing wave of light: potential $\propto \cos(2k_L x)$, so $\theta = 2k_L x$; momentum in units of $2\hbar k_L$.
- Kick strength $\kappa = 11.6$, scaled Planck constant $\bar{k} = 8\omega_r T = 2.0$.
- Diffusion at the classical rate $\kappa^2/2$, break after about 8 kicks, then exponentially localized (checked to 50 kicks).
- Their $\xi = \kappa^2/4\bar{k}^2 \approx 8.3$ is the $1/e$ point of the probability: half of our $\xi = D/\hbar^2$.
- They also saw the resonances, and the antiresonance sign flip in the first kicks.

# Conclusions

- Classical: diffusion with $D(K)$ oscillating around $K^2/2$; the Bessel expansion works to about 4%.
- Quantum: follows the classical growth up to $t^* \sim D/\hbar^2$, then stops.
- Momentum distribution: exponentially localized, $\xi \approx 1.2\, D/\hbar^2$ ($\pm 20\%$), following $D(K)$.
- Floquet theory explains the saturation; the Anderson mapping explains the localization.
- Unitarity versus chaos: quantum errors never grow exponentially.

**Open questions from this project**

- Why does $\xi\hbar^2/D$ wander in bands as $\hbar$ changes?
- What causes the narrow central peak of the averaged distribution?

# References


- G. Casati, B. V. Chirikov, F. M. Izraelev, J. Ford, in *Stochastic Behavior in Classical and Quantum Hamiltonian Systems*, Lect. Notes Phys. **93**, 334 (1979).
- B. V. Chirikov, Phys. Rep. **52**, 263 (1979).
- J. M. Greene, J. Math. Phys. **20**, 1183 (1979).
- A. B. Rechester and R. B. White, Phys. Rev. Lett. **44**, 1586 (1980).
- A. B. Rechester, M. N. Rosenbluth, R. B. White, Phys. Rev. A **23**, 2664 (1981).
- F. M. Izrailev and D. L. Shepelyansky, Theor. Math. Phys. **43**, 553 (1980).
- S. Fishman, D. R. Grempel, R. E. Prange, Phys. Rev. Lett. **49**, 509 (1982).
- D. L. Shepelyansky, Phys. Rev. Lett. **56**, 677 (1986).
- F. M. Izrailev, Phys. Rep. **196**, 299 (1990).
- F. L. Moore, J. C. Robinson, C. F. Bharucha, B. Sundaram, M. G. Raizen, Phys. Rev. Lett. **75**, 4598 (1995).
- F. Saif, Phys. Rep. **419**, 207 (2005).

# Backup: where the Bessel corrections come from

$D = K^2[C_0 + 2\sum_{j\geq 1} C_j]$ with $C_j = \langle \sin\theta_0 \sin\theta_j \rangle$ over the torus.

- $(\theta_0, \theta_1)$ are independent and uniform (the change of variables from $(\theta_0, p_0)$ is a shear), and $\theta_{j+1} = 2\theta_j - \theta_{j-1} + K\sin\theta_j$.
- Expand with $e^{iz\sin\theta} = \sum_l J_l(z) e^{il\theta}$ and keep the terms in which all angles cancel:

$$C_0 = \tfrac{1}{2}, \quad C_1 = 0, \quad C_2 = -\tfrac{1}{2}J_2(K), \quad C_3 = \tfrac{1}{2}[J_3^2(K) - J_1^2(K)]$$

- Lags up to 3 give $D/(K^2/2) = 1 - 2J_2 - 2J_1^2 + 2J_3^2$; the next order adds $2J_2^2$.
- Checked at $K = 22$: $C_2 = -0.0658$ (formula $-0.0657$), $C_3 = -0.0024$ ($-0.0025$).

# Backup: the quantum two-kick result

Heisenberg picture with $\hat{U} = F\hat{K}$: $p_2 = \hat{p} + K\sin\theta_0 + K\sin\theta_1$, and $\hat{p}|0\rangle = 0$.

- Through one free step the kick is weakened:

$$\hat{U}^\dagger \mathrm{e}^{in\theta}\hat{U} = \mathrm{e}^{i\hbar n^2/2}\,\mathrm{e}^{in\theta}\,\mathrm{e}^{iK_n\sin(\theta+n\hbar/2)}\,T^n, \qquad K_n = \frac{2K}{\hbar}\sin\frac{n\hbar}{2}$$

  with $T$ the shift of $\theta$ by $\hbar$.

- $\langle\sin^2\theta_1\rangle = \frac{1}{2}[1 - J_2(K_2)]$ and $\langle\{\sin\theta_0, \sin\theta_1\}\rangle = \cos\frac{\hbar}{2}\,[J_0(K_q) - J_2(K_q)]$.
- Limits: $\hbar \to 0$ classical; $\hbar = 4\pi$ gives $2K^2$ (resonance); $\hbar = 2\pi$ gives 0 (antiresonance).
- The FFT code reproduces it to $10^{-15}$.

# Backup: when the classical $D$ is ambiguous

- $K = 9.25$: the line $p = 0$ crosses an island that traps 9% of the matched ensemble forever.
- Matched ensemble: $D/(K^2/2) = 0.718$. Uniform torus ensemble: $0.800$. Both steady over 4000 kicks.
- Near the accelerator windows ($6.28 \leq K \leq 7.45$ for $l = 1$) diffusion is anomalous: at $K = 5.5$ the kurtosis of $\Delta p$ is about 300.
- So experiment 07 uses $K = 4.5$, 5, 9.75, 10.5 and 11, where both ensembles agree.